

Electron Thruster for Small Spacecraft

Ricardo Sebastián Casimiro 
contact: stm32f303ret6@gmail.com

Abstract

Atmospheric drag can limit the orbital lifetime of small spacecraft, while their size and power budgets constrain propulsion hardware. This paper investigates an electron thruster that uses the ionosphere to supply its return current, eliminating the need for stored propellant. A cathode biased below the spacecraft body emits an electron beam through an aperture; the body charges positively and collects ambient electrons to replenish the emitted charge. Particle-in-cell simulations (WarpX) of conducting bodies 10 mm in diameter predict 3.4-43 nN of beam thrust at 100-350 V, with beam-supply powers of 12-278 mW and escape fractions of 96-99 %. A model calibrated to these simulations and coupled to year-long orbital and ionospheric calculations estimates the power required for drag compensation. For a body 10 mm in diameter and 30 mm long in sun-synchronous orbits at 500-600 km, the estimated mean beam-supply demand is 10-46 mW near solar maximum and 0.4-2 mW near solar minimum. These mission estimates extrapolate the collection model beyond the simulated plasma conditions and exclude emitter and power-conversion overheads. The results support further investigation of electron propulsion for small spacecraft; extension to larger bodies requires validation of current-collection scaling and emitter integration.

1 Introduction

Conventional gridded ion thrusters require stored propellant, a feed system and a neutralizer [1]. This hardware adds mass, volume and integration complexity. This paper investigates a simpler architecture: emit electrons and let the ionosphere provide the return current.

The tradeoff is thrust per watt. For a directed exhaust, $F/P = 2\eta/v_e$, where η is the electrical-to-jet efficiency and v_e is the exhaust speed. Electrons accelerated through a few hundred volts reach high exhaust speeds, giving little momentum per unit energy. The simulated 200 V operating point delivers about 0.2 $\mu\text{N/W}$, making electron propulsion power-intensive at millinewton thrust levels.

At the nanonewton drag levels examined here, the beam-supply demand is milliwatts. A cathode, an aperture, a high-voltage supply and the spacecraft's conducting skin provide propulsion without stored propellant. The aim is to reduce the hardware needed for station keeping on small spacecraft; emitter and converter overheads remain part of the flight design.

2 Concept

A cathode held at $-V$ with respect to the spacecraft body emits electrons and accelerates them out through an aperture (Fig. 1). Each escaping electron leaves the body one charge more positive, so the body's conducting skin col-

lects ambient electrons from the plasma, and the supply returns them to the cathode. The body settles at the floating potential φ where collection equals emission. The beam is not recaptured: its electrons carry 100-300 eV, while φ is a few to a few tens of volts and the ambient electrons carry 0.1 eV. The collected electrons arrive from all directions and cancel 0.4-1.4 % of the beam thrust at 200-350 V.

3 Theory

The concept is a plain electrostatic accelerator.

Thrust is the momentum flux of the escaping beam:

$$F = \frac{I_{\text{esc}} \sqrt{2m_e \text{KE}}}{e}. \quad (1)$$

Here $I_{\text{esc}} = f_{\text{esc}}I$ is the escaping part of the emitted current; collimated exhaust is assumed.

The energy each electron actually leaves with is

$$\text{KE} = \kappa e (V - \varphi), \quad (2)$$

with KE in joules and V the supply voltage.

Jet power divided by electrical beam-supply power is

$$\eta = \frac{P_{\text{jet}}}{P} = \kappa \frac{V - \varphi}{V} f_{\text{esc}}. \quad (3)$$

Here $P = VI$; emitter and converter overheads are excluded.

The energy fraction κ reflects the simulated launch position; emission at cathode potential ideally gives $\kappa \approx 1$.

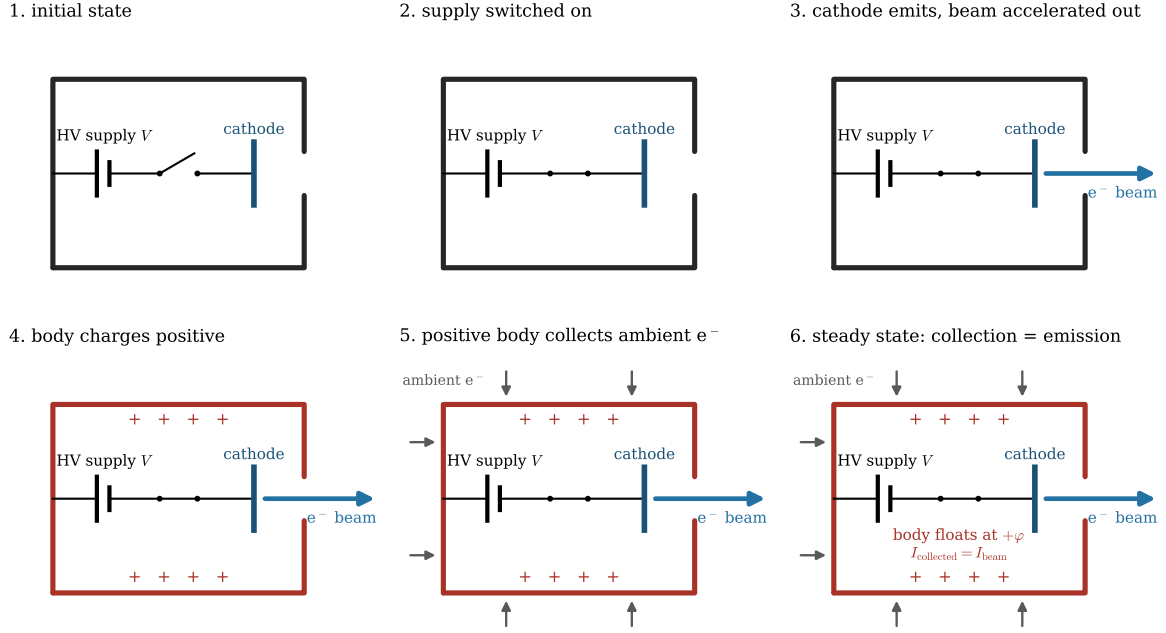


Figure 1: Operating sequence. The supply is switched on, the cathode emits, the escaping beam charges the body positive, and the positive body collects ambient electrons until collection equals emission.

The float potential φ depends on collecting area, body shape and plasma density. The “float tax” φ/V is the fraction of supply voltage lost to current return. The escape fraction f_{esc} is the beam fraction that clears the body, controlled by the exit-aperture geometry.

Control. Electrical measurements of current and floating potential can provide a calibrated thrust estimate. Feedback on the float can account for plasma density, temperature and day/night changes without an explicit ionosphere model.

4 Simulated capability

The device is simulated with WarpX [2, 3] in RZ electrostatic mode: a closed 10 mm can with a cathode disk inside and an aperture in the lid, in a dayside 400 km plasma ($n_e = 1.63 \times 10^{12} \text{ m}^{-3}$, $kT_e = 114 \text{ meV}$, ions at $400 m_e$), 0.15 mm cells, 800 ns runs. The body potential is not prescribed: each step it moves by the net charge that left and landed over the body’s self-capacitance. A ladder of nine stages builds from a vacuum electron gun to the full device, checking each piece against theory (thermal collection within 1 %); all pass, and every run passes fixed gates on escape, current balance and momentum. Table 1 gives the operating envelope. Two constants describe it: $F = 3.27 I \sqrt{KE}$ (with F in nN, I in mA and KE in eV; the slope includes interception and divergence) and $\kappa = 0.81$. Stretching the can to 30 mm ($L/r = 6$) multiplies its skin by 3.5 at the same frontal area: the float drops fourfold and thrust rises. A 3D run with the geomagnetic field across the beam at flight strength changes the thrust by under 1 %.

$\kappa = 0.81$ is a property of the simulation, not the device: the beam is launched 0.3 mm above the cathode,

Table 1: Measured operating envelope. $\eta = KE f_{\text{esc}}/eV$. Grid uncertainty: $F \pm 4\%$, $\varphi \pm 2\%$.

body	V V	I mA	φ V	F nN	P mW	f_{esc} %	η
compact	100	0.121	5.4	3.42	12.1	96.1	0.74
$10 \times 5.5 \text{ mm}$	200	0.342	17.0	13.65	68.4	98.4	0.73
	300	0.630	36.3	30.13	189	99.0	0.69
	350	0.793	48.3	40.48	278	99.1	0.68
slender	200	0.342	4.4	14.22	68.4	98.4	0.79
$10 \times 30 \text{ mm}$	350	0.793	14.0	43.33	278	99.1	0.77

where the potential is already 19 % of V above it, and the measured deficit is 17-18 % of V in every run. A real cathode emits at cathode potential ($\kappa \approx 1$), so every thrust and power figure here is conservative by about 11 %; a confirmation run is under way.

5 Station keeping

Drag and the plasma along the orbit come from a year-long TudatPy propagation [4] with NRLMSISE-00 density [5] driven by measured solar and geomagnetic indices and IRI-2020 plasma [6] along the orbit, near solar maximum (2024) and minimum (2019). The test body is the slender can of Table 1, 3.1 g, flown with its axis along the velocity, so the thrust axis and the low-drag pose coincide. With the thruster on, drag is cancelled exactly and the drag at each 5-minute point is the thrust demand; the supply voltage follows it at the least beam power, with the float estimated from the local plasma using the simulation-calibrated collection model. With the thruster off, the orbit decays freely.

Table 2 gives the result. Without the thruster the 3 g body re-enters from 400 km in 46 days and from 550 km

Table 2: Station keeping on a sun-synchronous orbit (10:30 ascending node). Each cell is 2024 (near solar maximum) / 2019 (near minimum). Drag is the year-mean thrust demand; lifetime is the free fall with the thruster off, to 120 km or the 5-year cap; power is the year-mean beam power to hold the altitude. The 1U drag is exact (drag scales with drag area, $\times 103$); its power is an estimate scaled the same way. [†]47 % of the points need more than 350 V: at the edge of the simulated envelope.

altitude	3 g slender can (10 $\times$ 30 mm)			1U CubeSat (1.33 kg), estimate		
	drag [nN]	lifetime, off	power [mW]	drag [μ N]	lifetime, off	power [W]
400 km	38 / 3.8	46 d / 241 d	278 [†] / 21	3.9 / 0.39	167 d / 2.4 yr	29 [†] / 2.2
500 km	8.6 / 0.49	204 d / 3.5 yr	46 / 2.0	0.89 / 0.050	2.1 yr / > 5 yr	4.8 / 0.20
550 km	4.3 / 0.21	329 d / 4.5 yr	20 / 0.78	0.45 / 0.021	> 5 yr / > 5 yr	2.1 / 0.080
600 km	2.3 / 0.10	2.2 yr / > 5 yr	9.6 / 0.37	0.23 / 0.010	> 5 yr / > 5 yr	0.98 / 0.039
650 km	1.2 / 0.057	> 5 yr / > 5 yr	4.8 / 0.21	0.12 / 0.006	> 5 yr / > 5 yr	0.49 / 0.022
700 km	0.66 / 0.037	> 5 yr / > 5 yr	2.5 / 0.14	0.068 / 0.004	> 5 yr / > 5 yr	0.26 / 0.014

in 11 months near solar maximum. With it, 500-700 km is held all year: the mean demand of 0.7-9 nN sits inside the measured envelope, for 2.5-46 mW near maximum and under 2 mW near minimum. At 400 km near maximum the 38 nN mean is below the 43 nN measured at 350 V, but the thin night-side plasma pushes half the points past 350 V; near minimum 400 km is held for 21 mW. The ISS and equatorial orbits differ from the sun-synchronous one by about 10 %.

The demand is not steady. It swings about $1.7\times$ over each orbit (day and night) and up to $2.5\text{-}2.8\times$ on storm days (11 May and 7-11 October 2024). Holding altitude is an impulse budget, so the peaks cost a small dip, not the orbit: replayed with thrust capped at the measured 350 V capability, the orbit never drops more than 10 m below its target at 500-700 km.

Drag, current and beam power all scale with the drag area, so the concept does not depend on size. A 1U CubeSat has 103 times the can’s drag area and needs 103 times its thrust and, to first order, its power: about 1 W at 600 km near solar maximum and tens of milliwatts near minimum. A 1U lives longer in free fall because its mass grows faster than its area: 167 days from 400 km near solar maximum, against 46 for the can.

6 Limitations

1. **Settling.** Ions are $400 m_e$ and floats are 800 ns snapshots; a 3D campaign settled its floats at 6μ s at about twice the 800 ns value, with thrust moving by a few percent.
2. **Prescribed beam.** Beam current is an input, launched above the cathode (κ above); the flight emitter is not chosen.
3. **CubeSat regime.** The simulated body is 2.5 Debye lengths across; a 1U is about 25, where the collection model has not been validated.
4. **Attitude and plasma flow.** A held pose is assumed and the plasma is stationary (no ram flow).

All powers are beam power VI ; converter and cathode

overheads belong to the flight design.

7 Conclusion

A cathode, a high-voltage supply and the spacecraft’s own skin make a thruster that needs no propellant. Simulations measure 3.4-43 nN from 12-278 mW, enough to hold a 3 g chip-scale spacecraft at 500-700 km for milliwatts where it would otherwise re-enter within a year, and the same physics scales with area to CubeSats at around a watt. The next steps are a settled, cathode-launched confirmation run and a large-body run in the CubeSat regime.

Data. Simulation inputs, run manifests, gate policies, orbit propagations and the model are in the project repository, archived at DOI:10.5281/zenodo.22115924.

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